Nasib Asgar

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EDUCATION

B.Sc in Naval Architecture and Marine Engineering, 04/2019 – 07/2024 | Dhaka, Bangladesh
Bangladesh University of Engineering and Technology (BUET)
CGPA of **3.26** out of **4.00**. (**3.50** in the final year)

MBA (in progress), 03/2025 – Expected 2027 | Dhaka, Bangladesh
Institute of Business Administration (IBA), University of Dhaka

RELEVANT COURSES

- Fluid Mechanics
- Marine Hydrodynamics
- Heat Transfer
- Basic Thermal Engineering
- Marine Engines and Fuels
- Optimization Methods

RESEARCH INTERESTS

Fluid dynamics
CFD, multiphase flow, hydrodynamics, aerodynamics.

Thermal sciences
Heat transfer, thermal management, combustion, thermodynamic efficiency.

Sustainable energy and materials
Alternative fuels, energy storage, materials for energy systems, decarbonization.

Data analysis and programming
Python pipelines, statistical modeling, numerical methods.

RESEARCH EXPERIENCE

Undergraduate Thesis
Title: Life Cycle Assessment of Alternative Marine Fuels for Inland Shipping of Bangladesh.
Supervisor: Professor Dr. N. M. Golam Zakaria, Department of NAME, BUET.
Description:

- Compared six alternative fuel systems (CNG, LNG, LPG, ethanol, hydrogen, battery electric) with diesel for an inland tanker on the Chattogram to Dhaka route.
- Ran a life cycle assessment in SimaPro and a life cycle cost analysis over 25 years. Battery electric came out best on both emissions and cost.
- Published at MARTEC 2024.
- Tools: SimaPro.

Capstone Project
Title: Designing a 148 TEU Inland Container Vessel.
Supervisor: Professor Dr. Md. Mashud Karim
Description:

- Designed a 148 TEU container vessel for the Pangaon to Chattogram route at 12 knots, in a group of three. The ship is 75.62 m long and 14.943 m wide, with a 3.78 m draft and 2,834 tonnes displacement.
- Took the design from principal particulars through hydrostatics, stability, Holtrop resistance (88.76 kN at 12 knots), propeller design and engine selection (two Yanmar 6AYEM-GT engines, 749 kW each). It meets every IMO intact stability criterion in all four loading conditions.
- Tools: AutoCAD, Rhinoceros 3D, Maxsurf.

PUBLICATIONS

Life Cycle Assessment of Alternative Marine Fuels for Inland Shipping of Bangladesh,
Jannatul Ambia Shorna, Nasib Asgar, N. M. Golam Zakaria
Proceedings of MARTEC 2024, International Conference on Marine Technology, Johor Bahru, Malaysia, September 2024, pp. 233–238.

Digital financial inclusion, institutional quality and income inequality (submitted; under review),
Md. Iftekharul Amin, K. M. Walid, Nasib Asgar (corresponding author), Alif Bin Hasan, Ittesaf Rahman
Panel of 36 developing economies, 2017–2022. I replicated the R analysis independently in Python and found two errors in our first draft.

PROJECTS

Heat-transfer simulations in COMSOL, *each one rebuilds a published benchmark*

- Natural convection in a differentially heated square cavity, Ra = 10^3 to 10^6. Nusselt number within 0.16% of the benchmark for the non-dimensional model and 0.34% for the dimensional one.
- Natural convection from a heated strip on a cavity floor, after Corvaro and Paroncini (2008). My Nu is 5 to 7% above the measured values, the same gap the original paper shows.
- Three mixed, conjugate and MHD convection cases with internal bodies, each checked against its source paper. One is within 0.45% of the benchmark.
- Tools: COMSOL.

NLP on 110,721 reviews of five Bangladeshi mobile finance apps, *MBA group project, IBA*

- Built the Python pipeline and led the analysis: review scraping, a two-model sentiment cascade, near-duplicate detection with MinHash LSH, bootstrap intervals, and a difference-in-differences analysis across 138 app releases.
- The model labelled 29.8% of reviews negative, or 17.0% after post-stratification weighting. Accuracy on a 600-review hand-labelled set was 66.5%.
- Tools: Python.

Present bias and unplanned spending on mobile financial services, *MBA group project, IBA*

- Surveyed 256 young urban users in two languages. Analyzed the data in Python with factor analysis, regression, moderation and mediation. Present bias predicted unplanned spending ($\beta = 0.309$, $p < .001$).
- Tools: Python.

TEST SCORES

IELTS score (September, 2024)
Overall: **7.5** (listening: 9, reading: 8.5, speaking: 7, writing: 6)

GRE Test Score (October 2024)
Total: **312** (Quant: 165, Verbal: 147, AWA: 3.5)

SKILLS

Python, MATLAB, C++, Fortran
Programming and numerical methods

COMSOL
Heat transfer simulations

Rhinoceros 3D
3D modeling

SimaPro
Life cycle assessment software

Abaqus
Finite element analysis

AutoCAD
2D drawing

Maxsurf
Hydrostatics and stability

pandas, SciPy
Regression and factor analysis

PROFESSIONAL EXPERIENCE

Industrial Attachment, *Chittagong Dry Dock Ltd*

- Spent a month in the yard learning welding, cutting, docking and mooring from experienced workers and senior engineers.
- Led a project on bringing Industry 4.0 to the dry dock, looking at where sensing and automation could fit existing processes.

CO-CURRICULAR ACTIVITIES

- Volunteered at the International Conference on Marine Technology (MARTEC), in 2022.
- Ex-member of the BUET Photography Club.
- Ex-member of the BUET Origami Club.